

Comparison: Churilov (arXiv:2605.17062v3) vs this repository's Campaign 4

2026-08-14 correction after direct bridge analysis: This earlier draft overstated agreement. The completed code-only rescore is authoritative: [CHURILOV_BRIDGE_ANALYSIS.md](#). Matching Churilov's extractor does **not** reconcile the rates; Campaign 4's bridge spans 8.30–17.49% under the dual-registry definition, above Churilov's 5.49–7.27% Python band. The original “range compression” and “reconciles exactly” statements below have been corrected accordingly.

Their paper: Aleksandr Churilov, *The Range Shrinks, the Threat Remains: Re-evaluating LLM Package Hallucinations on the 2026 Frontier-Model Cohort*. Independent researcher, April 2026. Artifact: [github.com/churik5/slopsquatting-replication-2026](#).

Our work: Campaign 4 of this repository ([REPLICATION.md](#) Part II, [campaign4_results.json](#)), August 2026.

Both replicate Spracklen et al. (USENIX Security '25) on a 2026 frontier cohort. We ran the comparisons below against our own committed data before writing this; the two cross-checks in §3 are reproducible from [Tests/](#) and the frozen master list.

1. The two studies at a glance

	Churilov	Campaign 4 (ours)
Cohort	Sonnet 4.6, Haiku 4.5, GPT-5.4-mini, Gemini 2.5 Pro, DeepSeek V3.2	opus-5, gpt-5.2, gpt-oss:20b, grok-4.6, deepseek-coder-v2:16b, deepseek-v4-flash
Languages	Python and JavaScript	Python only
Scale	Full corpus, 199,845 generations, no sampling	Sampled (200 prompts/dataset) + Track A replication with seeds
Extraction	Original regex extractor over generated code (imports + pip/npm install) + a noise filter	Grammar-aware Parser v2 over the two package-recommendation queries ; malformed as an outcome
Registry	Frozen 2024 master list only (current-snapshot revalidation "in progress")	Dual: frozen 2024 + current 2026-08 snapshot, plus stdlib exclusion
Statistics	Wilson binomial CIs, χ^2 + Holm, permutation robustness	Prompt- cluster bootstrap CIs, paired-by-prompt + Holm
Design	Post-hoc	Preregistered two-track
Beyond replication	Universal-hallucination set + coordinated disclosure (PyPI/Socket.dev); Jaccard matrix; refusal rates	Cap diagnostic; parser-contamination decomposition; Track A seed-variance
Cost	\$860.90	~ hosted API + local GPU

Cohorts are **near-disjoint** — same providers, adjacent versions — which is what makes the convergences below independent rather than shared-artifact.

2. Where we independently agree

- **A narrower selected cohort is not general range compression.** Churilov's five hosted models occupy a 4.62–6.10% overall band, but the comparison uses the original study's commercial/open-source category averages as endpoints rather than its model-level range. Campaign 4's purposive modern cohort spans 1.31–17.91% under a different recommendation estimand. Neither cohort is representative enough to establish frontier-wide compression.
- **Standard-library filtering is necessary.** Churilov filters stdlib (CPython list + Node `core_modules.csv`). We reached the same correction from the other side — the original Python pipeline *didn't* exclude stdlib, and we found squatters had registered stdlib names on 2024 PyPI (`os`, `sys`, `json`), so the original silently scored module-confusion as *valid packages*. Independent convergence on the same fix.
- **Registry staleness matters.** Churilov flags it as a limitation (§9.2) and says a current-snapshot revalidation "is in progress." **We already did it** — dual-registry scoring is committed, and our finding quantifies exactly their caveat: 8,961 names in the frozen 2024 list have since been *deleted* from PyPI, concentrated in the squat/spam class, so frozen-only scoring both over- and under-counts depending on the name. Their `git` example (blocked by PyPI's prohibited-name rules) and our stdlib-squat finding are the same phenomenon from two angles.
- **DeepSeek version pinning is broken.** Churilov's `deepseek-chat` alias returned metadata identifying `deepseek-v4-flash` — the model shifted under a stable alias. Our manifests were built to capture exactly this (served-model-id per response); we ran `deepseek-v4-flash` explicitly. Both studies independently document that DeepSeek's public alias is not a stable experimental identifier.
- **Reasoning-mode is a confound.** Churilov ran GPT-5.4-mini at `reasoning_effort=minimal` and got a 32% refusal rate, flagging the denominator asymmetry. We disabled reasoning where models exposed it and tracked empty/malformed as outcomes. Both grappled with the same reasoning-off confound; neither ran a full effort sweep.

3. Two cross-checks against our data

(a) Churilov's universal-hallucination set appears in our disjoint cohort. His central novel claim is a set of names all five of *his* models invent. We checked his top-8 universal PyPI names against our six (different) models:

Universal name	...also emitted by N of our 6 models
<code>objc</code>	5 (opus-5, gpt-5.2, grok-4.6, coder-v2, v4-flash)
<code>aws-cdk</code> , <code>opentelemetry</code>	2 each
<code>rest-framework</code> , <code>tencentcloud</code> , <code>mpl-toolkits</code> , <code>opengl</code> , <code>openstack</code>	1 each

All 8 appear in at least one of our models. This independently corroborates the model-agnostic universal-set claim with a cohort that shares no exact model with his — stronger evidence for his finding than his single-cohort data alone can provide.

(b) The direct code-only bridge does not reconcile the rates. We initially treated our Q1 package query as a proxy for Churilov's import extractor. That was incorrect. We have now run the actual H1-plus-import bridge on the frozen generated code:

Campaign 4 model	Churilov-compatible frozen rate	Dual-registry/non-stdlib rate
Claude Opus 5	9.52%	9.52%
GPT-5.2	8.30%	8.30%
gpt-oss 20B	10.74%	10.61%
Grok 4.6	18.15%	17.49%
DeepSeek Coder V2	9.13%	8.87%
DeepSeek V4 Flash	8.89%	8.89%
Qwen 3.5 Cloud (separate extension)	7.10%	6.51%
Kimi K2.7 Code Cloud (separate extension)	7.38%	7.38%
Kimi K3 Cloud (successor extension)	8.62%	8.37%

The three later extension cells sharpen the picture. Qwen's dual-registry bridge rate falls inside Churilov's reported 5.49–7.27% Python band, and Kimi K2.7 sits 0.11 points above its upper edge; Kimi K3 scores 8.37%. There is still no exact model overlap, and the original six-cell bridge remains 8.30–17.49%, so this is cohort-sensitive partial convergence rather than numerical reconciliation. The Kimi package-query aggregates are different estimands dominated by extreme Query 2 enumeration and must not be compared directly with Churilov's code imports.

Thus extraction channel alone is not the explanation. Of the original six cells' 490 dual-registry bridge flags, 477 arise from imports and only 13 from explicit install directives. Synthetic-prompt cells are also much higher than Stack Overflow cells. The remaining live explanations include genuine cohort behavior, prompt/source composition, stochastic sampling, and the semantic weakness of mapping an import module directly to a PyPI distribution or treating a local project module as an external dependency.

4. Where each study is ahead

Churilov has, and we do not:

- **Both languages**, and the Python-over-JavaScript inversion (Spracklen's 2024 pattern was JS-worse) — a real finding we cannot speak to.
- **Scale**: the full 199,845-generation corpus, not a sample.
- **Real security follow-through**: the universal set → coordinated disclosure with PyPI Security and Socket.dev → a registrable attack surface of 53 names. Genuinely novel and operationally valuable; we did no cross-model intersection.
- **A full-corpus Jaccard overlap matrix** and the DeepSeek/GPT-5.4-mini training-origin hint.
- A named cohort of **current commercial frontier** models; ours mixed in local/open-weight and some idiosyncratic picks.

We have, and Churilov does not:

- **Preregistration** with logged amendments.
- **Prompt-cluster CIs** instead of Wilson binomial — see §5.
- **Dual-registry + stdlib categories** (the revalidation he defers).
- The **cap diagnostic** and the **parser-contamination decomposition** — the reason our band sits lower and is, we argue, cleaner.
- **Concentration / prompt-level-risk split**, which separates a model's typical behavior from its catastrophic tail (§3b). His pooled rate would hide such a tail — though his code-import extraction is far less tail-prone than our Q2, so the risk is smaller for his design.
- **Prompt-conditioned overlap analysis** over names and aligned `(name, dataset, prompt-index)` occurrences: five names span all six original Track B cells, but four have same-prompt support across every cell. This directly tests shared elicitation as an alternative to a training-origin explanation.
- **Two independent adversarial audits** and a verified-not-asserted correction trail.

5. One specific, checkable caution on his statistics

Churilov's per-model CIs are **Wilson intervals over the multiset of extracted references** (§4.6), treating each reference as an independent Bernoulli trial. References cluster within generations — one code file emits several imports whose validity is correlated — so the effective sample size is smaller than $|R|$ and the Wilson intervals are **too narrow**. This is the exact understatement the audit of *our* work caught, and that our first Phase 5 pass committed: an unstratified/unclustered interval made an Opus-vs-GPT difference look significant that did not survive prompt-clustered resampling.

His headline pair (Haiku 4.5 vs GPT-5.4-mini, $p \approx 4 \times 10^{-12}$) is extreme enough to survive any reasonable clustering correction. But the **marginal** surviving pairs — Sonnet 4.6 vs GPT-5.4-mini at Holm- $p \approx 6 \times 10^{-3}$ — are the ones most likely to weaken under generation-clustered resampling, and "5 of 10 pairs significant" may be optimistic. A cluster bootstrap over generations (the fix in our `compare_to_paper.py`) would settle it, and would be a cheap addition to his artifact requiring no new model runs.

6. Bottom line

The studies are complementary but do not generally numerically reconcile. Churilov is broader in languages, corpus size, registrability testing, and disclosure; Campaign 4 is stronger on parser validation, dual registries, prompt-cluster inference, and failure shape. The direct bridge sharpens the original six-cell disagreement, while Qwen 3.5's later extension cell lands inside Churilov's Python band, Kimi K2.7 lands just above it, and Kimi K3 reaches 8.37%. That cohort-sensitive partial convergence identifies cohort composition and import semantics as joint next validation targets. Prompt-conditioned overlap also qualifies the shared-training interpretation: five names span all six Campaign 4 models, but four are emitted by every model on an identical prompt. The defensible synthesis is therefore not "the frontier compressed," but "package-hallucination rates remain highly measurement- and corpus-dependent while the cross-model attack surface is operationally real."

Two concrete things we could offer his effort: the completed current-registry revalidation he defers to future work, and a generation-clustered re-computation of his pairwise significance.

Prepared by Claude (Fable 5) for the repository maintainer; the §3 cross-checks were computed against this repository's committed Campaign 4 data and are reproducible from `Tests/`.